

VINEETH KAIMAL

Technical Leader | Power Integrity, Thermal & Packaging

Nokia Finland | Ex-ANSYS | Ex-Qualcomm

ABOUT ME

Results-driven engineering leader with 15 years of expertise in Power Integrity, Advanced Packaging, SoC PDN sign-off, and Thermal Analysis. Demonstrated success in building and directing global cross-functional teams (25+ engineers), setting technical strategy, and delivering AI-driven automation platforms that reduce silicon area, lower cost, and ensure product reliability across cutting-edge technology nodes.

PROFESSIONAL EXPERIENCE (MASTER LIST)



CONTACTS

- +358 405490885
- vineethkaimal1989@gmail.com
- Oulu, Finland
- linkedin.com/in/vineeth-kaimal-373707a8

EDUCATION

B.E. (HONS) ELECTRICAL & ELECTRONICS ENGINEERING

2007 – 2011 | CGPA: 7.26/10
BITS Pilani, K.K. Birla Goa Campus

HIGHER SECONDARY (+2)

2007 | Score: 600/600 (100%)
H.S.E, Kerala

10TH STANDARD (SECONDARY)

2005 | Score: 87.6%
C.B.S.E Board

SKILLS & EXPERTISE

LEADERSHIP

- Global Team Direction (25+)
- Technical Strategy
- Stakeholder Management

ENGINEERING DOMAINS

- Power Integrity (PI)
- Package Thermal
- SoC PDN Sign-off
- IR Drop / EM / ESD
- CAD Methodology

AUTOMATION & SCRIPTING

- Python
- Perl
- Tcl
- Shell Scripting
- HTML & CSS
- JavaScript / jQuery

KEY TOOLS

- Redhawk
- Redhawk-SC
- Redhawk-SC Electrothermal
- Totem
- System PI
- Voltus
- Power Artist

TECHNICAL LEADER – POWER INTEGRITY Dec 2022 – Present

Nokia • System-level PI, Advanced Packaging & Thermal Leadership

- Define and own technical strategy for system-level power integrity and package thermal sign-off across high-speed product portfolios.
- Architect and lead development of AI-driven automation platforms for package ball visualization, layout modification, and ballout optimization – directly reducing design cycle time.
- Drive SoC die size optimization from a PI perspective, delivering measurable silicon area and cost reduction.
- Lead advanced package-level and component thermal analysis programs to proactively mitigate reliability risks.

STAFF / LEAD / SENIOR ENGINEER Jan 2016 – Dec 2022

Qualcomm • SoC & Hard-Macro PDN Sign-off | Team Lead

- Built, mentored, and directed a 25-member global engineering team delivering 6+ concurrent sign-off programs across 3lpe, 4ff, 5lpe, and 7ff nodes.
- Owned end-to-end SoC PDN sign-off strategy; drove early floorplan architecture, padding layout, bump planning, and ESD placement decisions.
- Successfully led the Hard-Macro level Power Delivery Network (PDN) sign-off team to ensure PDN convergence is seamless with best-in-class quality metrics.
- Provided feedback about PDN health to HM PD implementation engineers from floorplan stage to Metal Tape-Out.
- Engineered 200+ automation scripts (Perl/Tcl) and an automated HTML reporting system, unifying sign-off visibility across all hard macros.
- Leveraged ANSYS RedHawk, RedHawk-SC, and Cadence Voltus for static/dynamic IR drop, signal EM, and ESD clamp-to-instance validation.
- Liaisoned between HM implementation teams and SoC Convergence team, preventing late-stage tape-out issues.

SENIOR ENGINEER Aug 2015 – Jan 2016

INVECAS (via Soctronics) • CAD and Methodology Team

- Established internal CAD methodologies and automated PI validation sign-off flows adopted across active production projects.

APPLICATIONS ENGINEER Aug 2011 – Aug 2015

ANSYS Apache • Technical Interface for Tier-1 Accounts

- Served as primary engineering interface for Tier-1 accounts including Qualcomm (Bangalore), Atheros (Shanghai), STMicroelectronics, ST Ericsson, and Ericsson.
- Supported RedHawk (SoC level power noise/reliability) and Totem (Transistor level power noise/reliability sign-off).
- Drove customer engagements for new solutions, preparing training presentations, conducting workshops, and delivering custom feature prototyping for development teams.

INTERNSHIPS

SOFTWARE QUALITY ASSURANCE (SQA) INTERN May 2011 – Aug 2011

Broadcom India Research Pvt. Ltd.

- Developed Perl scripts for automating the documentation of Linux kernel testing results.

IC DESIGN FRONT END TEAM INTERN July 2010 – Dec 2010

STMicroelectronics (Gr. Noida, India)

- Worked with the Central CAD & Infrastructure (CCI) IC design front-end team.

ENGINEERING INTERN

May 2009 – July 2009

BHEL (Combined Cycle Demonstration Plant) (Trichy, India)

- Exposure to thermal power generation plant methodology and combined cycle systems.

ENGINEERING PROJECTS

OPTIMAL PLACEMENT OF WIND [MATLAB](#), [Monte-Carlo Simulation](#) TURBINES IN A WIND FARM

Implemented Monte-Carlo simulation to determine the optimum placement of wind rotors in a wind farm considering wake effects.

WIND ANALYSIS USING WEIBULL [WEBA Software](#), [MATLAB](#) DISTRIBUTION

Fitted wind velocity profiles with a Weibull distribution to evaluate site wind power capacity.

COMMUNICATION SYSTEMS SIGNAL [MATLAB](#) MULTIPLEXING

Designed a circuit model to multiplex 5 audio signals, perform FM modulation, and retrieve original signals without noise.

BASIC IC TESTER APPLICATION [8086 Microprocessor Assembly](#)

Designed and tested IC validation logic using 8086 microprocessor assembly.

COMPONENT V-I CHARACTERISTICS [8051 Microcontroller](#) PLOTTER

Plotted V-I characteristics of electronic components on a graphical LCD using 8051 microcontroller.

HONORS & ACHIEVEMENTS

- 100% Score (600/600) in Kerala Higher Secondary Board Examination (2007).
- Awarded Merit-Cum-Need Scholarship at BITS-Pilani, K.K. Birla Goa Campus.
- Runner-up in CODE WARS (programming event) at QUARK '09 (Annual Technical Festival of BITS Pilani).
- Finalist in Electrade (circuit design event) and participant in SYMPHONY (MATLAB event) at QUARK '10.

EXTRA-CURRICULAR ACTIVITIES & LEADERSHIP

- 5-time Kerala State Chess Champion; represented Kerala in National Chess Championship 15 times; Peak FIDE Rating: ELO 2014.
- Elected Mess Secretary at BITS Pilani Goa Campus (2009–2010); directed a team of 14 student representatives in menu planning, hygiene standards, brand sourcing, billing, and software maintenance.
- Core Committee Member for QUARK '09 & '10 technical festivals, leading artist invitations and event management.

- Member of BITS Pilani Goa Campus Badminton Team.

